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High Performance Processors and Systems

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Name
Last Name

Problem 1 (50%)	
Problem 2 (50%)	
Total (100%)	

Problem 1

1. To implement cache coherency one of the protocols is the MESI concurrency protocol implemented for a write-invalidate write-back cache. Describe briefly the algorithm (also in term of graph) and complete the following table.

State	Cache up to date?	Memory up to date?	Others have a copy?	Cache can respond other's reads?
Modified				
Exclusive				
Shared				
Invalid				

2. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid (0)	Yes	Yes
1	P2: read block 0				
2	P1: write block 0				
3	P2: read block 2				
4	P1: read block 1				
5	P2: write block 1				
6	P1: read block 1				
7	P2: write block 1				
8	P1: read block 0				
9	P2: write block 1				
10	P2: write block 1				
11	P1: read block 1				
11	P2: write block 1				
12	P1: read block 1				
13	P2: read block 1				

Problem 2

Define the performance measures, CPU time and MIPS. Why is CPU time a better performance measure than MIPS? How do you summarize performance when comparing between two alternatives over a set of benchmarks?

